Reinforcement Learning from Human Feedback

Training language models to follow instructions

Language models predict the next token

Given n tokens, predict token $n+1$.

$$p_\theta(w_{n+1} \mid w_1, w_2, \ldots, w_n)$$

The cat sat on the ___

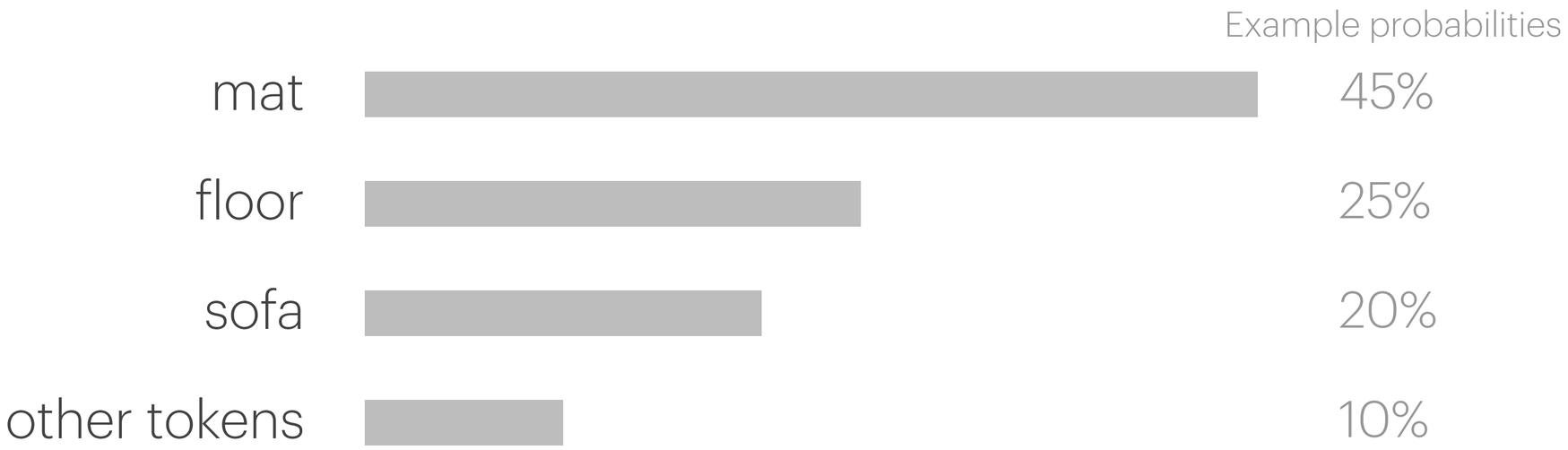
θ represents the model parameters.

Tokens can be words or parts of words.

Learn a probability distribution

The model assigns a probability to each possible next token.

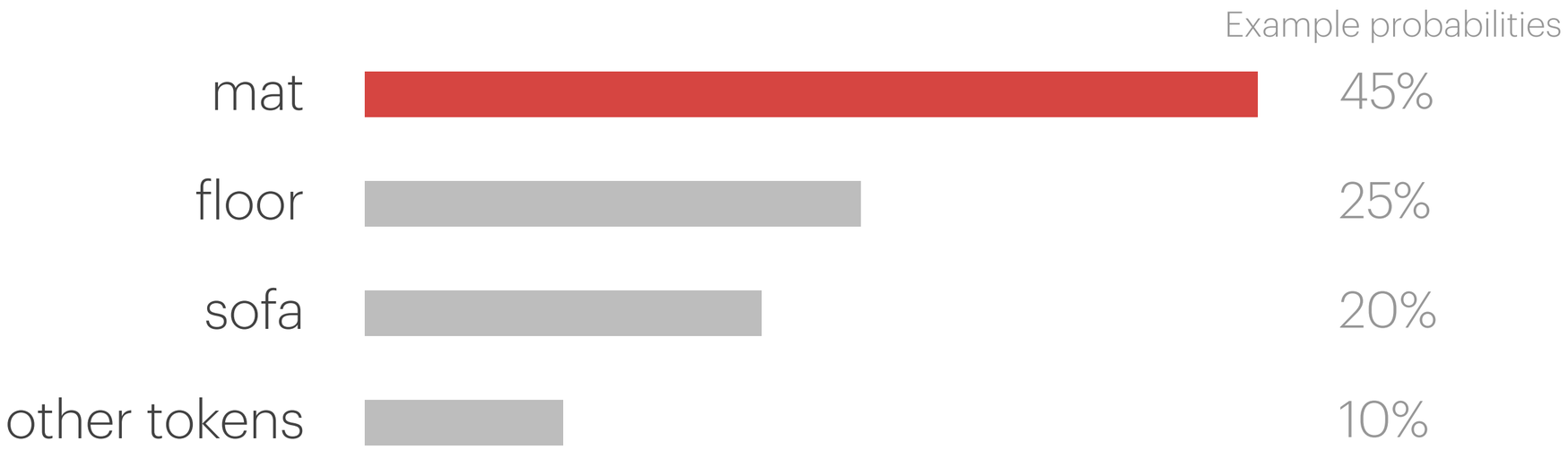
The cat sat on the ___



Learn a probability distribution

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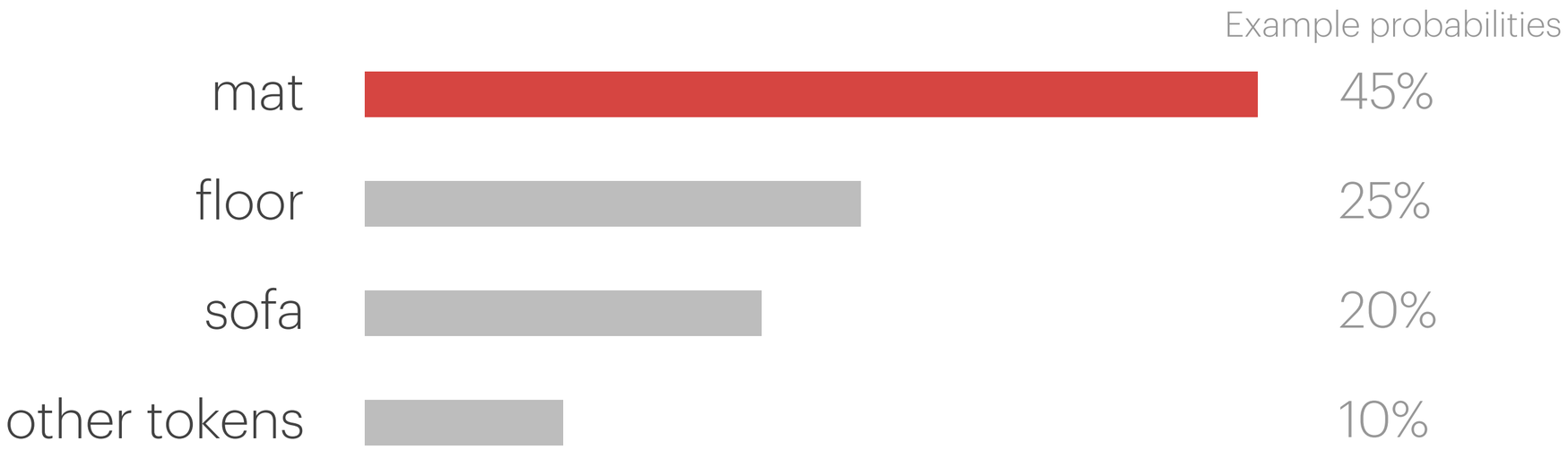


The next token in this training example is “mat”.

Learn a probability distribution

The model assigns a probability to each possible next token.

The cat sat on the ___



The next token in this training example is “mat”.

Training raises the probability of the target token.

Generate text one token at a time

Sample a token. Add it to the context. Repeat.

The cat sat on the

Generate text one token at a time

Sample a token. Add it to the context. Repeat.

The cat sat on the

The cat sat on the mat

Generate text one token at a time

Sample a token. Add it to the context. Repeat.

The cat sat on the

The cat sat on the mat

The cat sat on the mat .

Sampling does not always select the most likely token.

We want LLM output to be good to read

Q Explain overfitting to a beginner.

A1 The model learns the training examples too closely.
It then performs poorly on new examples.

A2 Overfitting occurs when empirical risk minimization
yields poor out-of-sample generalization.

Illustrative responses

We want LLM output to be good to read

Q Explain overfitting to a beginner.

A1 The model learns the training examples too closely.
It then performs poorly on new examples.

A2 Overfitting occurs when empirical risk minimization yields poor out-of-sample generalization.

Both answers describe overfitting.
A1 is easier for a beginner to understand.

Illustrative responses

The training target matters

Predicting likely text does not always complete the user's task

Pretraining	User's task
Predict likely continuations	Produce a useful response

The training target matters

Predicting likely text does not always complete the user's task

Pretraining	User's task
Predict likely continuations	Produce a useful response

A bigger model does not automatically change the target.

Three training stages

Demonstrations

Train the language model

Supervised fine-tuning
(SFT)

Three training stages

Demonstrations

Train the language model

Supervised fine-tuning (SFT)

Comparisons

Train a reward model

Reward modeling

Three training stages

Demonstrations	Comparisons	Policy updates
Train the language model	Train a reward model	Improve its responses
Supervised fine-tuning (SFT)	Reward modeling	Reinforcement learning (RL)

Two kinds of human feedback

Write an example answer

Supervised fine-tuning

Compare possible answers

Two kinds of human feedback

Write an example answer

Supervised fine-tuning

Compare possible answers

Reward modeling

Several answers can be correct. People may still prefer one.

Humans rank complete responses

Rank these answers for a beginner who is learning about overfitting.

A The model learns the training examples too closely. It then performs poorly on new examples.

B Overfitting occurs when empirical risk minimization yields poor out-of-sample generalization.

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C The model fits its training data too closely
and does not generalize well.

D The predictor fits sample-specific noise,
reducing empirical risk without reducing population risk.

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Rank these answers for a beginner who is learning about overfitting.

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D The predictor fits sample-specific noise,
reducing empirical risk without reducing population risk.

One possible ranking: A > C > B > D

Human preference as a reward signal

For the same prompt, we want to measure how much people prefer A to B.

Answer A

Answer B

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$$P(A \succ B \mid x)$$

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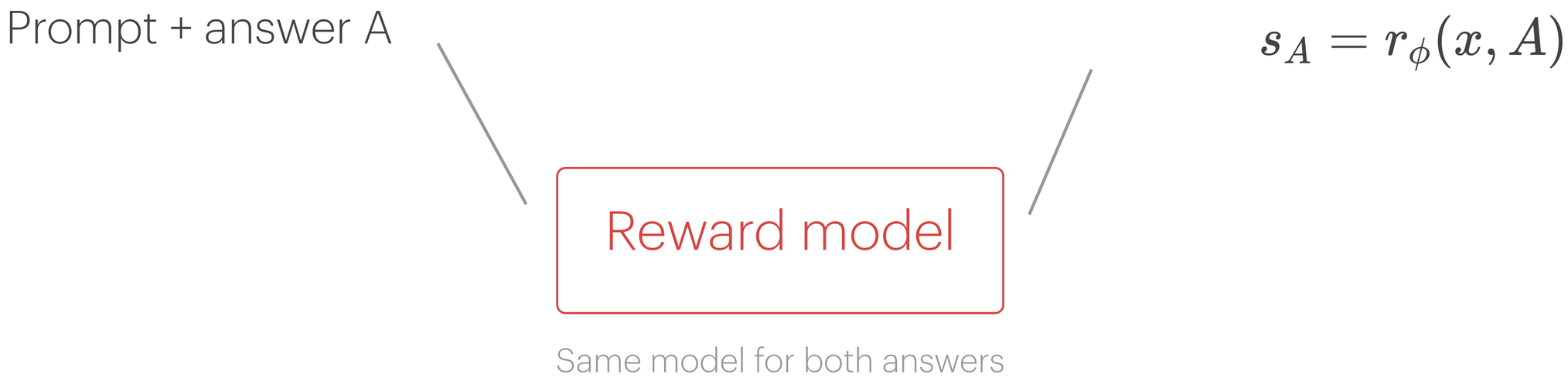
0.51: slight preference for A

0.99: strong preference for A

The signal should increase with preference for A.

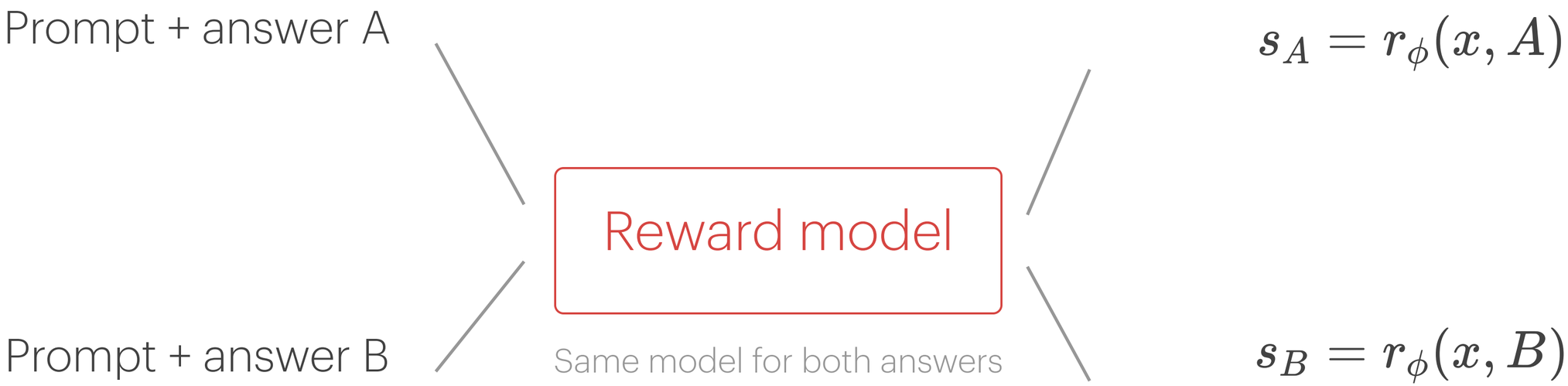
Learn scores for both answers

We construct this preference signal with one reward model.



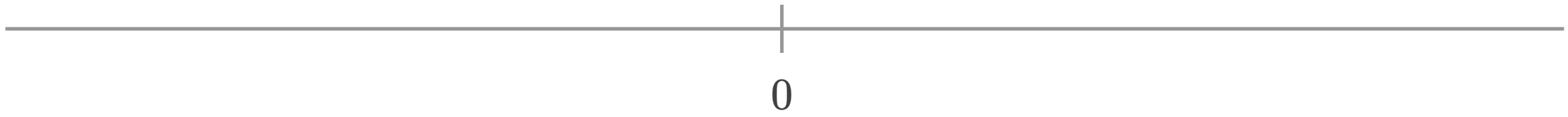
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Compare the scores

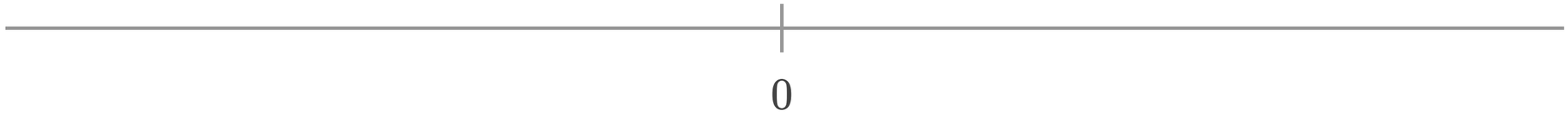
$$\Delta = s_A - s_B$$



Compare the scores

$$\Delta = s_A - s_B$$

$\Delta < 0$ $\Delta > 0$



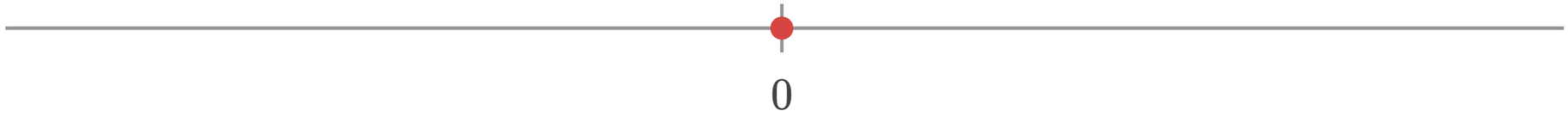
The model predicts a preference for B.

The model predicts a preference for A.

Compare the scores

$$\Delta = s_A - s_B$$

$\Delta < 0$ $\Delta > 0$

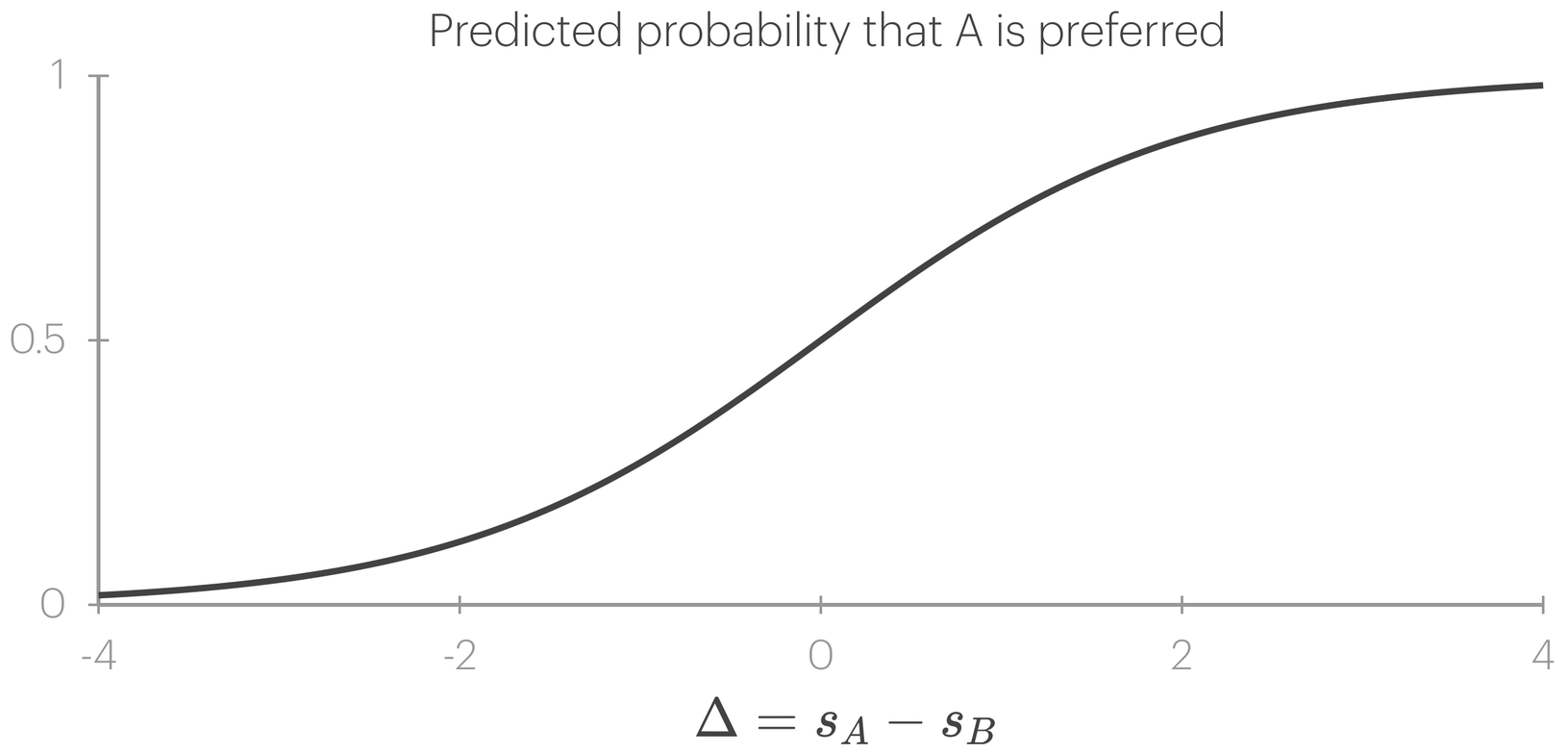


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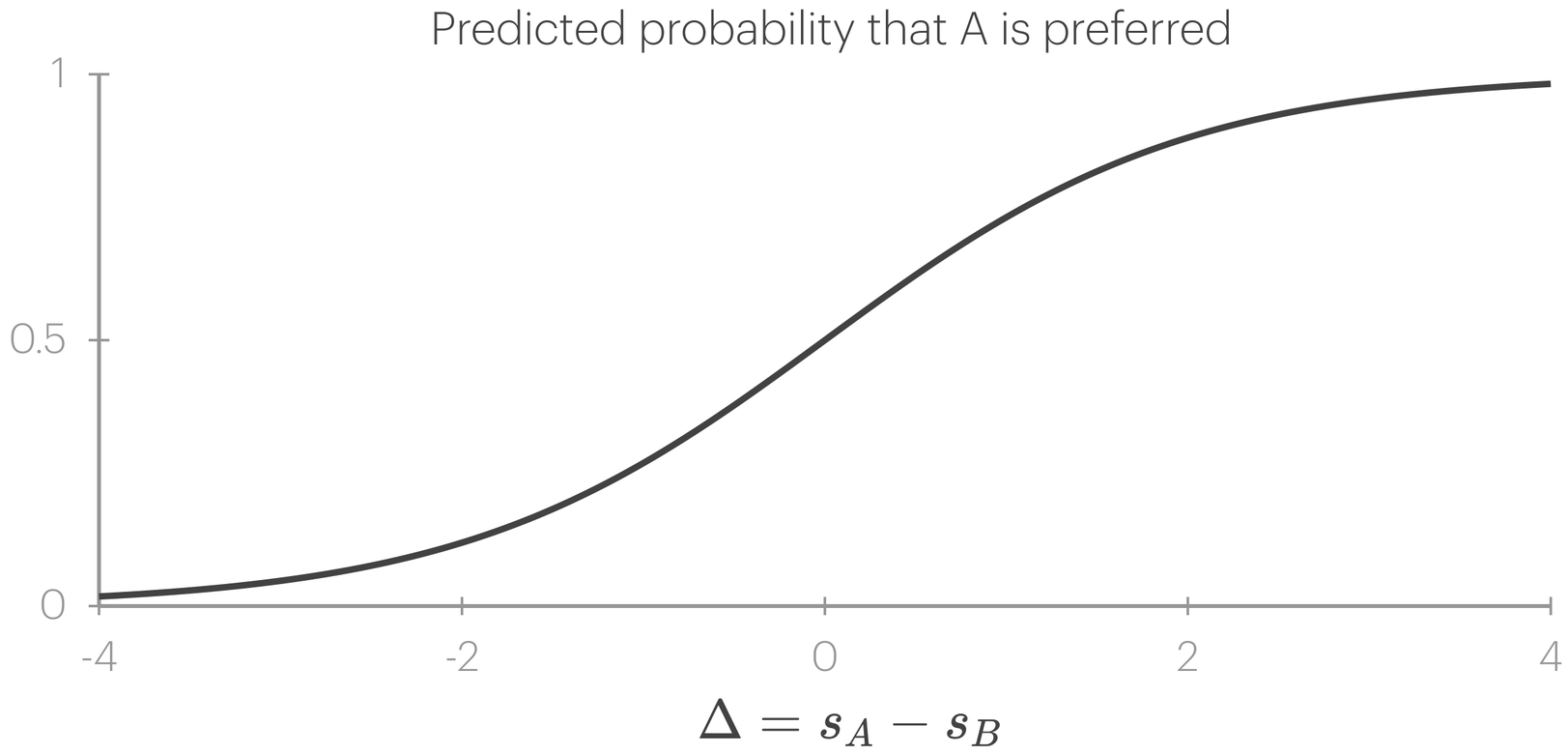
Zero means equal scores.

Predict the preference probability



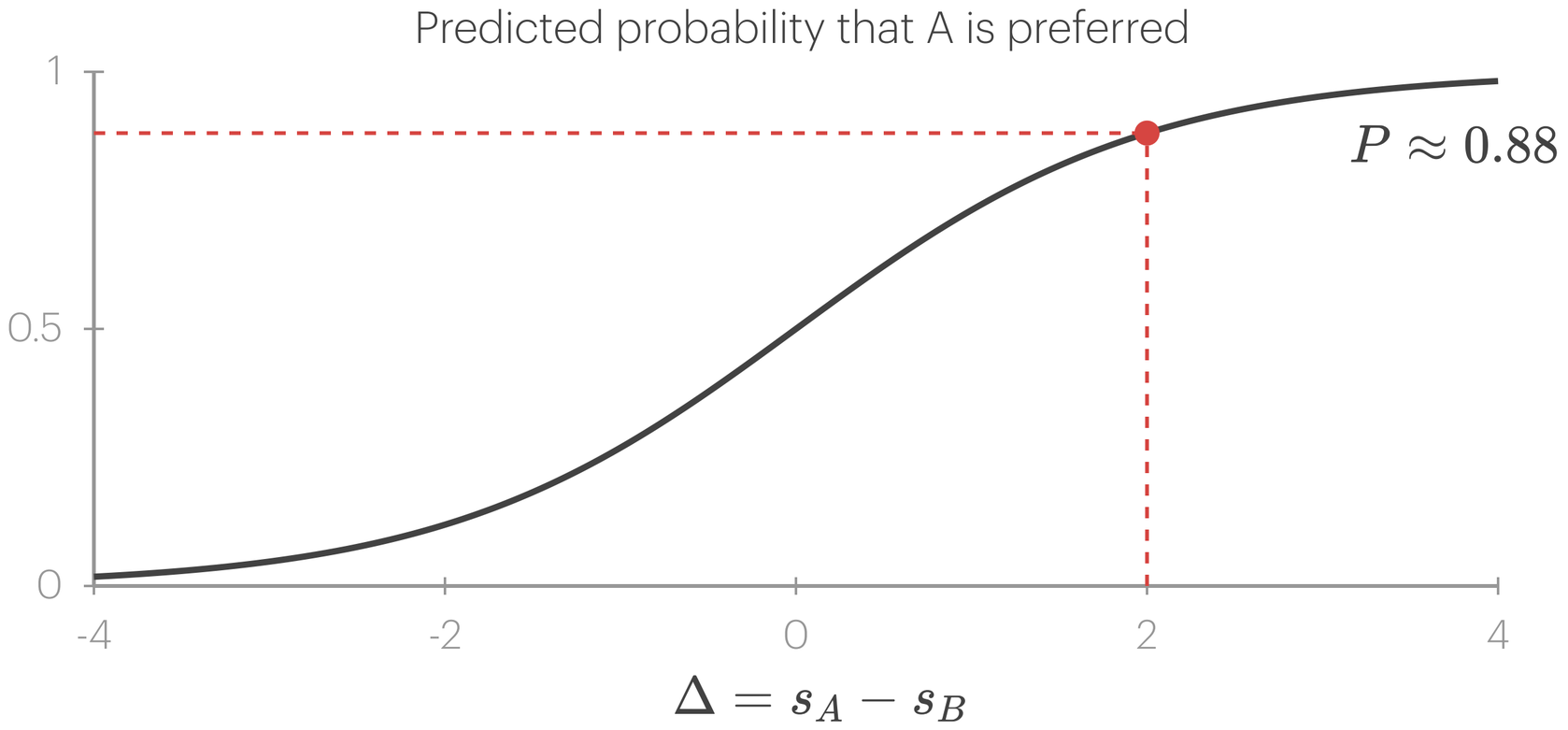
Predict the preference probability

$$P(A \succ B \mid x) = \mathbf{sigmoid}(\Delta)$$



Predict the preference probability

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The example scores give an 88% predicted preference for A.

Learn from the human comparison

The person preferred A to B.

$$\mathcal{L} = -\log P(A \succ B \mid x)$$

Reinforcement Learning on the LLM

Two models

Policy π_θ

Generates a response

Updated

Reward model

Scores the response

Not updated

Two models

Policy π_θ

Generates a response

Updated

Reward model

Scores the response

Not updated

$$R(x, y) = r_\phi(x, y)$$

One training iteration

The reward model stays fixed. The policy changes.

Sample

Generate a response

One training iteration

The reward model stays fixed. The policy changes.

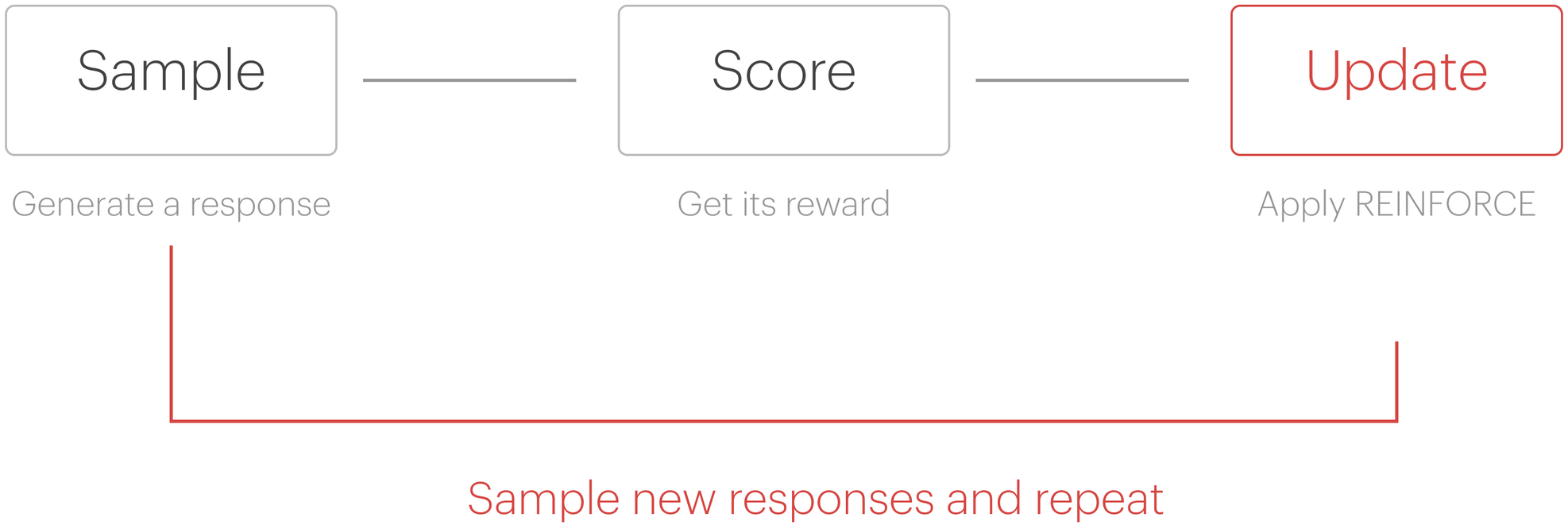


Generate a response

Get its reward

One training iteration

The reward model stays fixed. The policy changes.



High reward can still produce poor answers

A possible failure when the reward model makes a mistake.

Repeated updates can exploit errors in the reward model.

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Possible response: “Overfitting is always good.
A model should memorize every example.”

Reward model: high score

High reward can still produce poor answers

A possible failure when the reward model makes a mistake.

Repeated updates can exploit errors in the reward model.

Possible response: “Overfitting is always good.
A model should memorize every example.”

Reward model: high score

The policy can move far from its starting behavior.
Answers can become false or nonsensical.

Keep a copy of the starting policy

The reference stays fixed while the policy learns.



Generates a response

Updated

Scores the response

Not updated

Keep a copy of the starting policy

The reference stays fixed while the policy learns.

Policy π_θ

Generates a response

Updated

Reward model

Scores the response

Not updated

$$\pi_{\text{ref}} = \pi_{\theta_0}$$

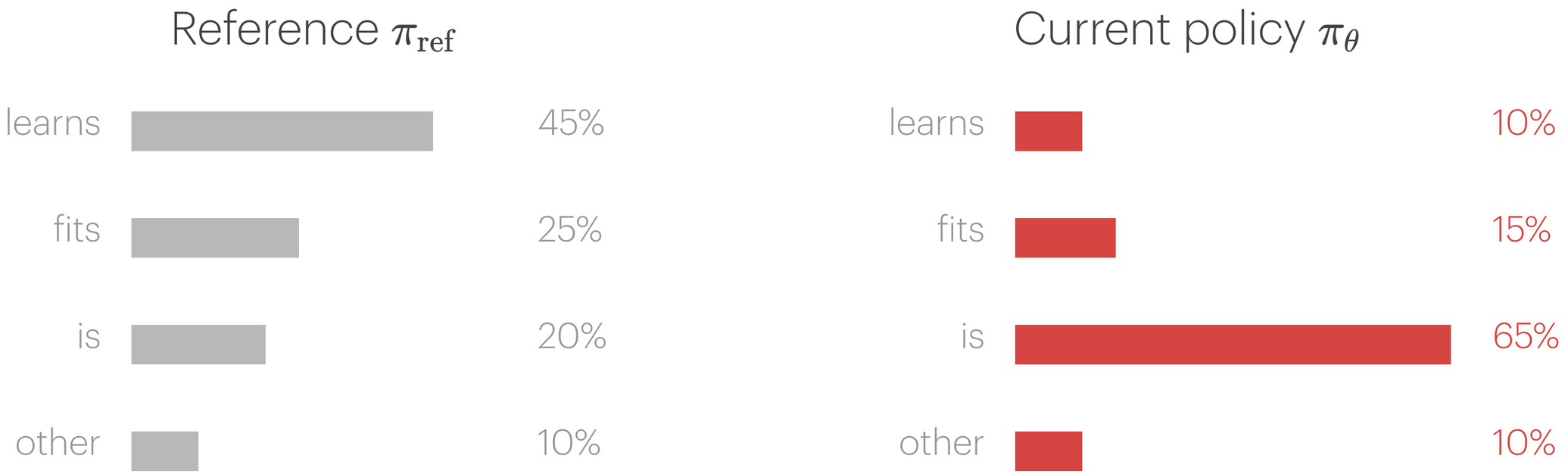
Reference π_{ref}

Copy made before any RL updates

Not updated

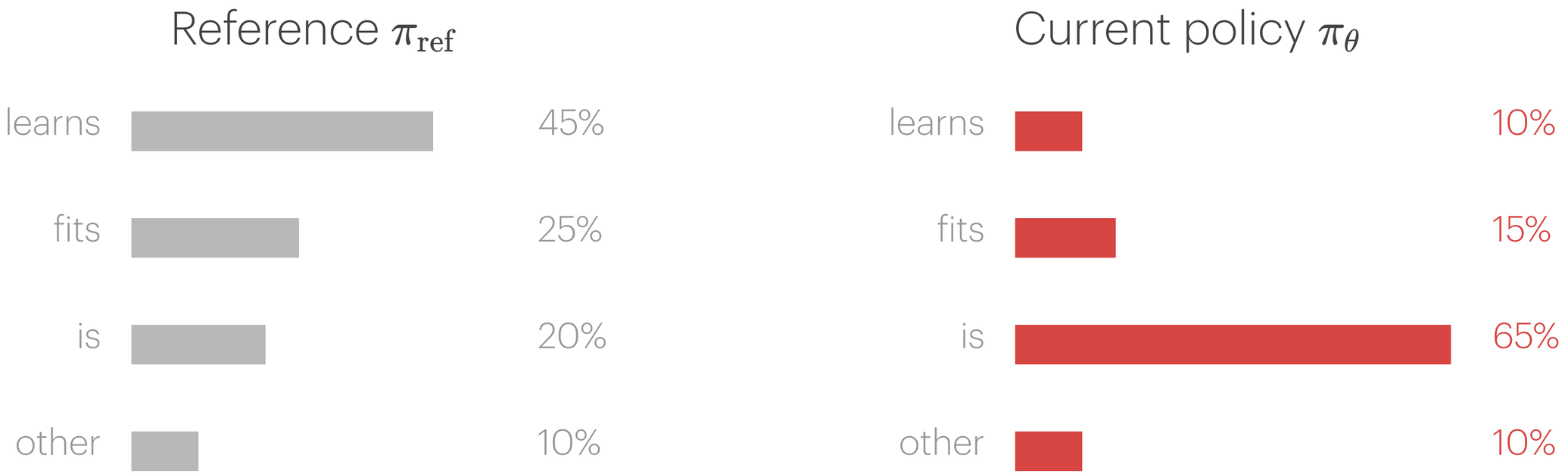
Compare the two distributions

Example next-token probabilities after “The model”.



Compare the two distributions

Example next-token probabilities after “The model”.



Penalize large changes from the initial distribution.

KL divergence

KL divergence measures how one probability distribution differs from another.

$$D_{\mathrm{KL}}(P\|Q) = \sum_{w} P(w) \log \frac{P(w)}{Q(w)}$$

KL divergence

KL divergence measures how one probability distribution differs from another.

$$D_{\mathrm{KL}}(P\|Q) = \sum_{w} P(w) \log \frac{P(w)}{Q(w)}$$

It is zero when the distributions match. It is not symmetric.

It is widely used to compare probability models and regularize training.

Add the KL penalty

$$\textbf{Training objective} = \textbf{expected score} - \beta\,\textbf{KL divergence}$$

Add the KL penalty

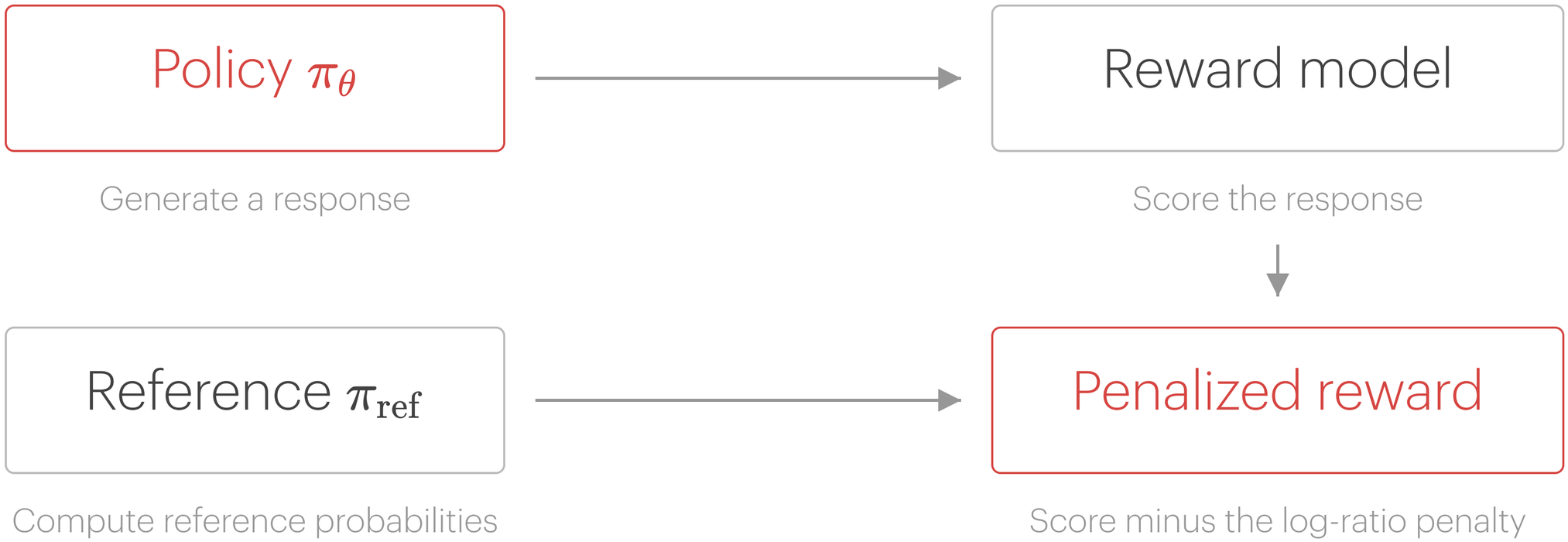
$$\textbf{Training objective} = \textbf{expected score} - \beta\,\textbf{KL divergence}$$

$$R(x, y) = r_\phi(x, y) - \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

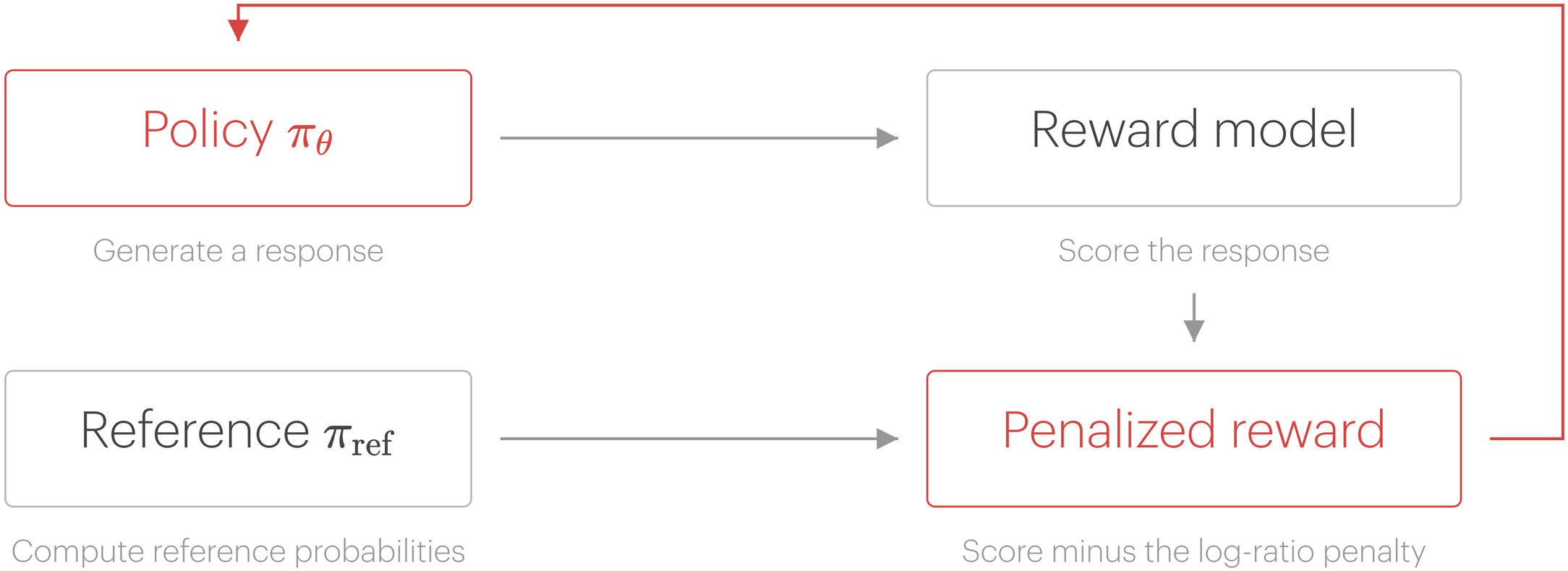
The expected log-ratio penalty is KL divergence.

β controls the penalty for changes from the reference.

The updated training loop



The updated training loop



REINFORCE updates the policy using the penalized reward.

The objective with KL

Fix one prompt x and maximize the expected reward.

$$R_\theta(x, y) = r_\phi(x, y) - \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

The objective with KL

Fix one prompt x and maximize the expected reward.

$$R_\theta(x, y) = r_\phi(x, y) - \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

$$J(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)}[R_\theta(x, y)]$$

The reward model and reference stay fixed. Only the policy changes.

Differentiate the objective

Start from J

$$J(\theta) = \sum_y \pi_\theta(y \mid x) R_\theta(x, y)$$

$$\nabla_\theta J = \nabla_\theta \sum_y \pi_\theta(y \mid x) R_\theta(x, y)$$

Differentiate the expected reward.

Differentiate the objective

Differentiate the sum

$$\nabla_\theta J = \nabla_\theta \sum_y \pi_\theta(y \mid x) R_\theta(x, y)$$

$$\boldsymbol{\nabla_\theta J = \sum_y \nabla_\theta [\pi_\theta(y \mid x) R_\theta(x, y)]}$$

Move the gradient inside the sum.

Differentiate the objective

Apply the product rule

$$\nabla_\theta J = \sum_y \nabla_\theta [\pi_\theta(y \mid x) R_\theta(x, y)]$$

$$\nabla_\theta J = \sum_y [R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x) + \pi_\theta(y \mid x) \nabla_\theta R_\theta(x, y)]$$

Use the product rule. Both factors depend on θ.

Differentiate the objective

Separate the terms

$$\nabla_\theta J = \sum_y [R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x) + \pi_\theta(y \mid x) \nabla_\theta R_\theta(x, y)]$$

$$\nabla_\theta J = \sum_y R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x) + \sum_y \pi_\theta(y \mid x) \nabla_\theta R_\theta(x, y)$$

Separate the two sums.

Differentiate the objective

Differentiate the reward

$$\nabla_\theta J = \sum_y R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x) + \sum_y \pi_\theta(y \mid x) \nabla_\theta R_\theta(x, y)$$

$$\nabla_\theta R_\theta = \nabla_\theta \left[r_\phi(x, y) - \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)} \right]$$

Now compute the reward derivative in the second sum.

Differentiate the objective

Expand the log ratio

$$\nabla_\theta R_\theta = \nabla_\theta \left[r_\phi(x, y) - \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)} \right]$$

$$\nabla_\theta R_\theta = \nabla_\theta r_\phi - \beta \nabla_\theta [\log \pi_\theta - \log \pi_{\text{ref}}]$$

Expand the log of the ratio. The prompt and response stay fixed.

Differentiate the objective

Differentiate the fixed models

$$\nabla_\theta R_\theta = \nabla_\theta r_\phi - \beta \nabla_\theta [\log \pi_\theta - \log \pi_{\text{ref}}]$$

$$\nabla_\theta R_\theta = 0 - \beta [\nabla_\theta \log \pi_\theta - 0]$$

The reward model and reference do not depend on θ.

Differentiate the objective

Simplify the reward derivative

$$\nabla_\theta R_\theta = 0 - \beta[\nabla_\theta \log \pi_\theta - 0]$$

$$\boldsymbol{\nabla_\theta R_\theta = -\beta \nabla_\theta \log \pi_\theta(y \mid x)}$$

Only the current-policy term remains.

Differentiate the objective

Differentiate log π

$$\nabla_\theta R_\theta = -\beta \nabla_\theta \log \pi_\theta(y \mid x)$$

$$\nabla_\theta \log \pi_\theta(y \mid x) = \frac{\nabla_\theta \pi_\theta(y \mid x)}{\pi_\theta(y \mid x)}$$

Use the derivative of a logarithm.

Differentiate the objective

Substitute back into ∇J

$$\nabla_\theta \log \pi_\theta(y \mid x) = \frac{\nabla_\theta \pi_\theta(y \mid x)}{\pi_\theta(y \mid x)}$$

$$\nabla_\theta J = \sum_y R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x)$$
$$- \beta \sum_y \pi_\theta(y \mid x) \frac{\nabla_\theta \pi_\theta(y \mid x)}{\pi_\theta(y \mid x)}$$

Substitute the reward derivative into the two-sum expression.

Differentiate the objective

Cancel the probability factors

$$\nabla_\theta J = \sum_y R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x) - \beta \sum_y \pi_\theta(y \mid x) \frac{\nabla_\theta \pi_\theta(y \mid x)}{\pi_\theta(y \mid x)}$$

$$\nabla_\theta J = \sum_y R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x) - \beta \sum_y \nabla_\theta \pi_\theta(y \mid x)$$

Cancel π in the numerator and denominator of the second sum.

Differentiate the objective

Simplify the second sum

$$\nabla_\theta J = \sum_y R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x) - \beta \sum_y \nabla_\theta \pi_\theta(y \mid x)$$

$$-\beta \sum_y \nabla_\theta \pi_\theta(y \mid x) = -\beta \nabla_\theta \sum_y \pi_\theta(y \mid x)$$

Move the gradient outside the second sum.

Differentiate the objective

Use probability normalization

$$-\beta \sum_y \nabla_\theta \pi_\theta(y \mid x) = -\beta \nabla_\theta \sum_y \pi_\theta(y \mid x)$$

$$-\beta \nabla_\theta \sum_y \pi_\theta(y \mid x) = -\beta \nabla_\theta 1$$

The probabilities of all responses sum to one.

Differentiate the objective

Differentiate the constant

$$-\beta \nabla_\theta \sum_y \pi_\theta(y \mid x) = -\beta \nabla_\theta 1$$

$$-\beta \nabla_\theta 1 = 0$$

The derivative of a constant is zero.

Differentiate the objective

Return to the gradient

$$-\beta \nabla_\theta 1 = 0$$

$$\nabla_\theta J = \sum_y R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x)$$

The second sum is zero. Keep the first sum.

Differentiate the objective

Rewrite the first sum

$$\nabla_\theta J = \sum_y R_\theta(x, y) \nabla_\theta \pi_\theta(y \mid x)$$

$$\nabla_\theta J = \sum_y \pi_\theta(y \mid x) R_\theta(x, y) \frac{\nabla_\theta \pi_\theta(y \mid x)}{\pi_\theta(y \mid x)}$$

Multiply and divide each term by π.

Differentiate the objective

Write the log-probability gradient

$$\nabla_\theta J = \sum_y \pi_\theta(y \mid x) R_\theta(x, y) \frac{\nabla_\theta \pi_\theta(y \mid x)}{\pi_\theta(y \mid x)}$$

$$\nabla_\theta J = \sum_y \pi_\theta(y \mid x) R_\theta(x, y) \nabla_\theta \log \pi_\theta(y \mid x)$$

Use the log-derivative identity again.

Differentiate the objective

The REINFORCE form

$$\nabla_\theta J = \sum_y \pi_\theta(y \mid x) R_\theta(x, y) \nabla_\theta \log \pi_\theta(y \mid x)$$

$$\boldsymbol{\nabla_\theta J = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)}[R_\theta(x, y) \nabla_\theta \log \pi_\theta(y \mid x)]}$$

A probability-weighted sum is an expectation.

The sampled REINFORCE update

Sample fresh responses. R includes the reward score and the KL penalty.

$$\hat{g} = \frac{1}{N} \sum_{i=1}^{N} R_\theta(x, y_i) \sum_{t=1}^{T_i} \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t})$$

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$$\theta \leftarrow \theta + \alpha \hat{g}$$

Use each sampled return as a fixed multiplier of the log-probability gradient.

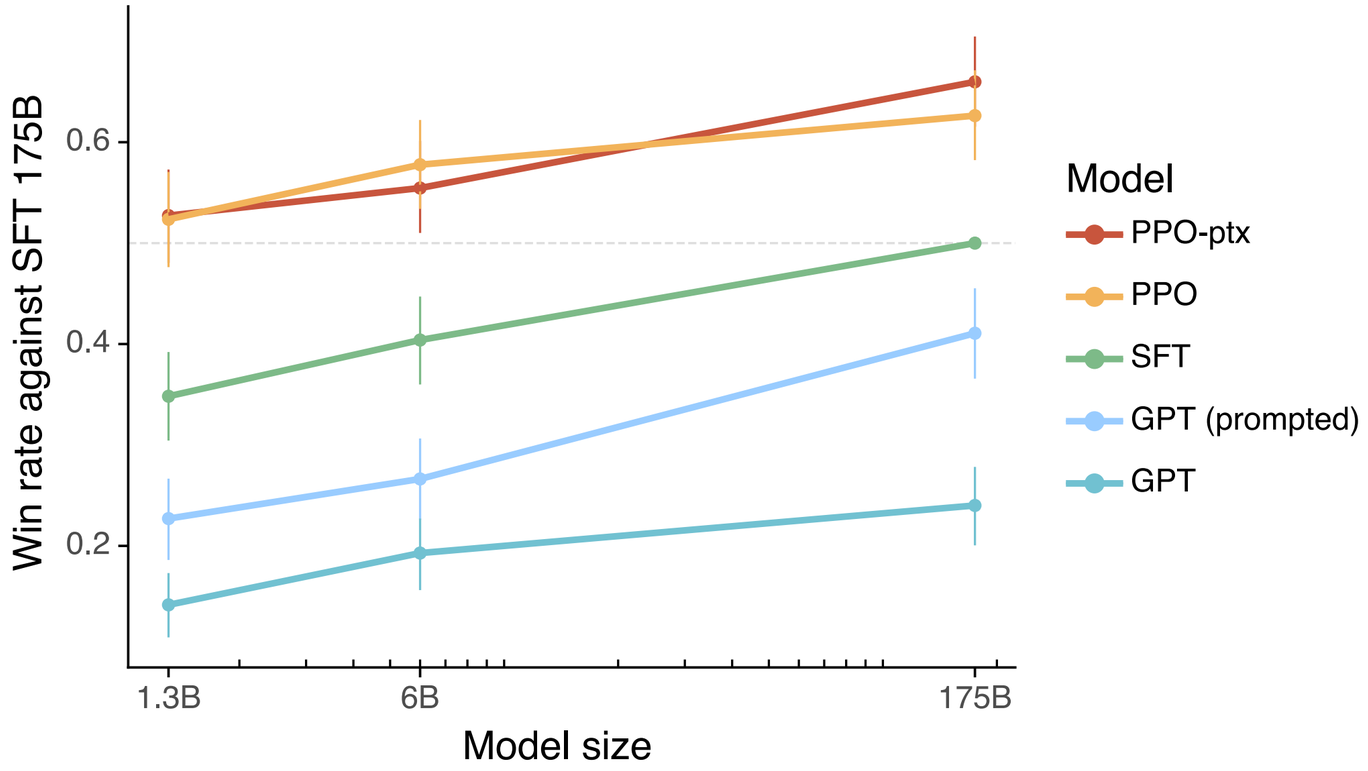
InstructGPT

OpenAI · 2022

Fine-tuning GPT-3 with human feedback

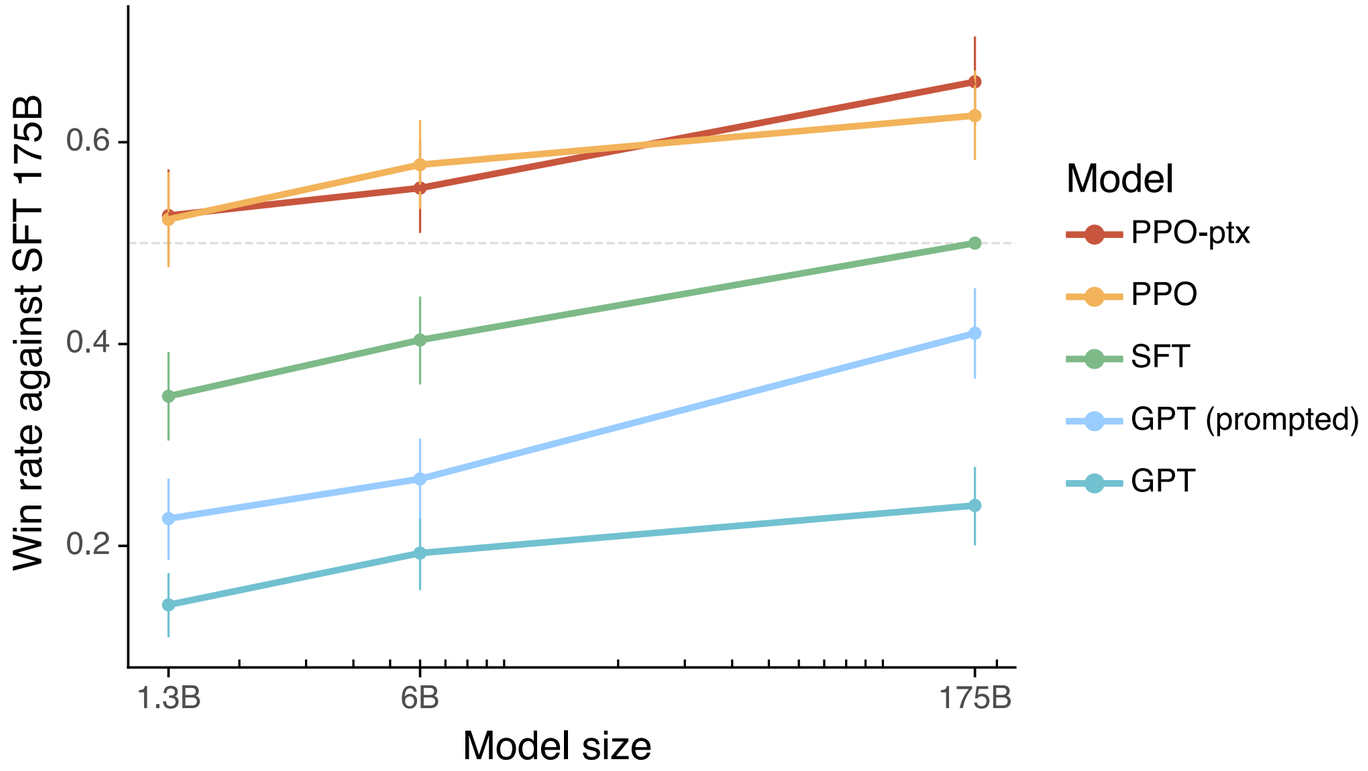
Main preference result

Preference rates against the 175B SFT model



Main preference result

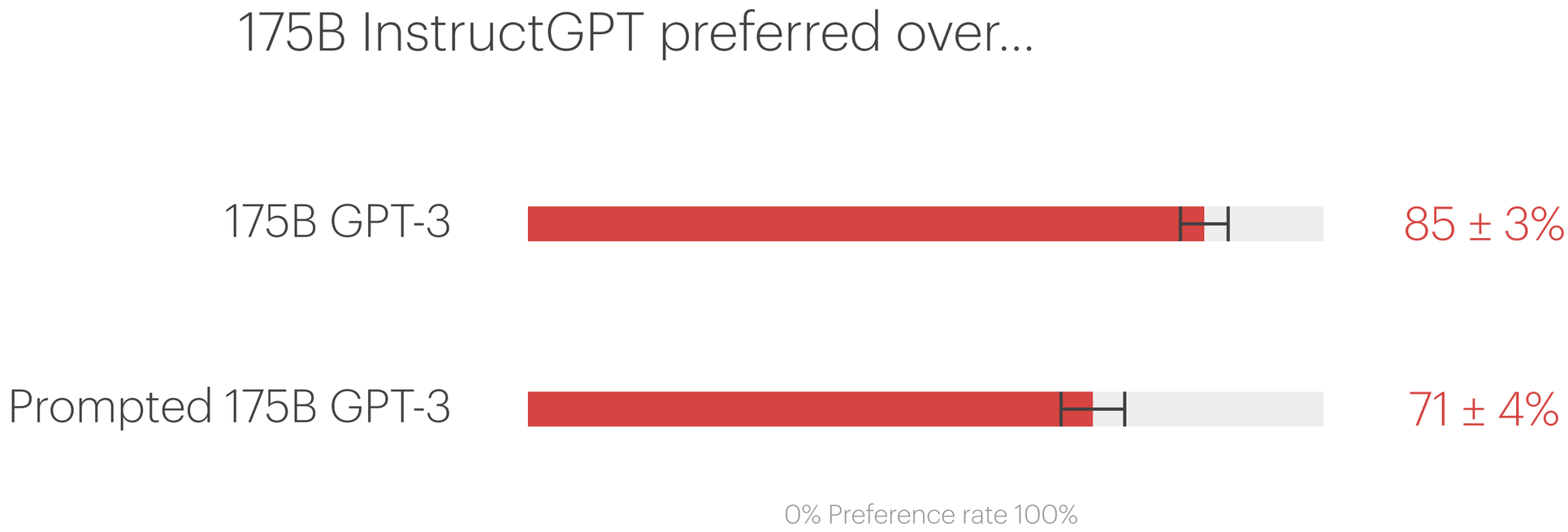
Preference rates against the 175B SFT model



1.3B InstructGPT is preferred to 175B GPT-3 on this evaluation.

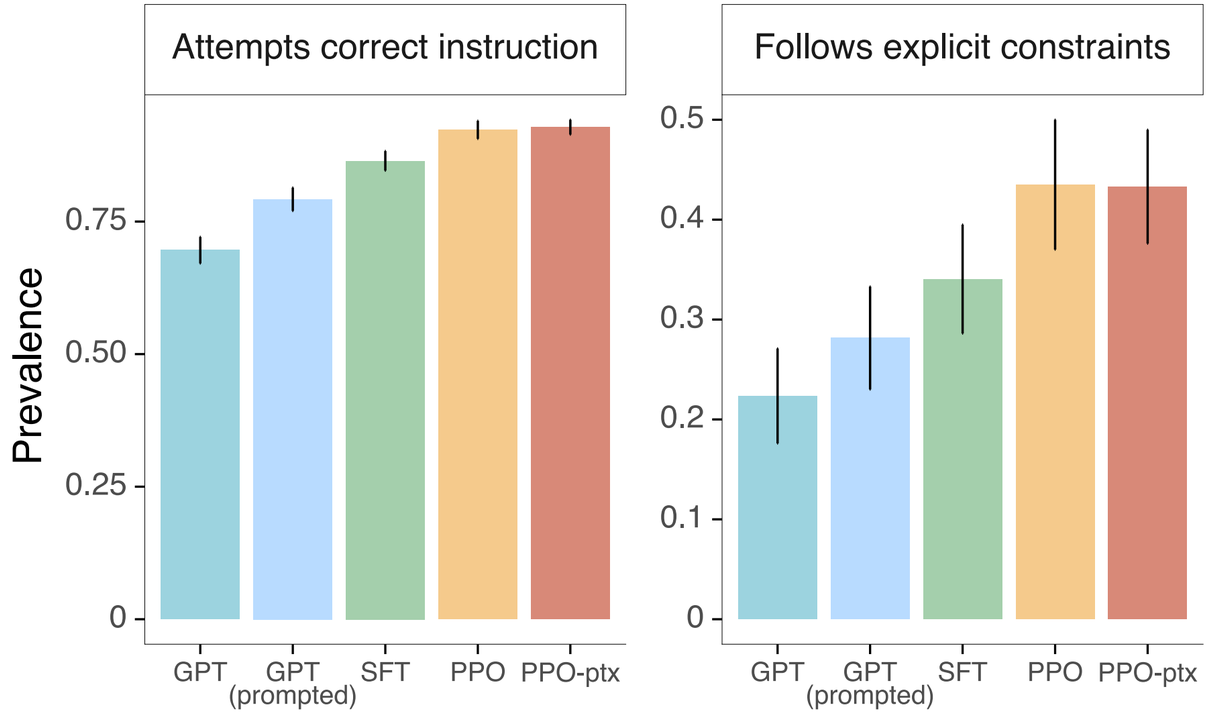
Direct comparisons of 175B models

A separate comparison from the previous figure



Changes in response behavior

Task completion and instruction following



InstructGPT more often attempts the task and follows the stated constraints.

Changes in response behavior

Task completion and instruction following

Added facts that were not in the supplied text

41%

GPT-3

21%

InstructGPT

A narrower measure than “all factual errors.”

Training from human feedback

The model learns from example answers and human comparisons.

Explain overfitting to a beginner.

Demonstrate	Train the language model
Compare	Train a reward model
Optimize	Improve the policy. Penalize changes from the reference.

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A higher reward does not always mean a more useful answer.
The reward model can misrepresent human preferences.